TE2 Worked Examples: Verifying Trigonometric Identities

Recording companion for the TE2 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What does this side look like in sines and cosines? Almost every TE2 verification starts the same way: rewrite every function in terms of sine and cosine. Then distribute, cancel what multiplies, combine fractions with a common denominator, and use $\sin^2 + \cos^2 = 1$ if it appears. Finish by renaming 1/cos as sec, cos/sin as cot, and so on, until you reach the other side. Start from either side, usually the more complicated one, and make each line equal the one before. Do not do the same operation to both sides.

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Example 1: Rewrite 1

Table 1. Tangent, cotangent, secant, and cosecant written in terms of sine and cosine

tan x	sin x / cos x
cot x	cos x / sin x
sec x	1 / cos x
csc x	1 / sin x

$$\tan x \csc x$$

Question

Which single function equals this product?

- **A.** csc x
- **B.** sec x
- **C.** $(\sin^2 x)/(\cos x)$
- **D.** tan x sec x

Answer and reasoning

tan x csc $x=(\sin x)/(\cos x)\cdot 1/(\sin x)=1/(\cos x)=\sec x$. The sines cancel because they multiply. The answer tan x sec x mixes up cosecant and secant, using csc $x=1/(\cos x)$; the answer $(\sin^2 x)/(\cos x)$ uses csc $x=\sin x$, and csc x comes from canceling the cosine instead of the sine. **tan x csc $x=\sec x$.**

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Example 2: Rewrite 2

$$\sec x \csc x$$

Question

Which expression equals this product?

- **A.** $\sin x \cos x$
- **B.** $1/(\sin x+\cos x)$
- **C.** $1/(\sin x \cos x)$
- **D.** $(\cos x)/(\sin x)$

Answer and reasoning

$\sec x \csc x=1/(\cos x)\cdot 1/(\sin x)=1/(\sin x \cos x)$. Multiplying fractions multiplies the denominators; it does not add them. The answer $\sin x \cos x$ forgets that these are reciprocals. **$\sec x \csc x=1/(\sin x \cos x)$.** This product shows up in many CheckIt identities.

Example 3: Rewrite 3

$$\cot x \sec x$$

Question

Which single function equals this product?

- **A.** $\csc x$
- **B.** $\sec x$
- **C.** $(\cos^2 x)/(\sin x)$
- **D.** $\tan x$

Answer and reasoning

$\cot x \sec x = (\cos x)/(\sin x) \cdot 1/(\cos x) = 1/(\sin x) = \csc x$. The answer $(\cos^2 x)/(\sin x)$ uses $\sec x = \cos x$, and $\tan x$ flips the cotangent instead of simplifying. **$\cot x \sec x = \csc x$.**

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Example 4: Rewrite 4

$$\frac{\tan x}{\sec x}$$

Question

Which single function equals this quotient?

- **A.** $\cos x$
- **B.** $\frac{\sin x}{\cos^2 x}$
- **C.** 1
- **D.** $\sin x$

Answer and reasoning

Dividing by $\sec x = \frac{1}{\cos x}$ is the same as multiplying by $\cos x$: $\frac{\sin x}{\cos x} \cdot \cos x = \sin x$. The answer $\frac{\sin x}{\cos^2 x}$ multiplies by $\sec x$ instead of dividing, and 1 cancels everything. **$\frac{\tan x}{\sec x} = \sin x$.**

Example 5: Simplify 1

$$\tan x\,(\sin x+\cos x)$$

Question

Which expression equals this one?

- **A.** $(\sin^2 x)/(\cos x)+\sin x$
- **B.** $\sin^2 x+\sin x$
- **C.** $(\sin^2 x)/(\cos x)+1$
- **D.** $(\sin x)/(\cos x)\cdot\sin x+\cos x$

Answer and reasoning

Distribute: $\tan x \sin x+\tan x \cos x=(\sin^2 x)/(\cos x)+(\sin x)/(\cos x)\cdot\cos x=(\sin^2 x)/(\cos x)+\sin x$. CheckIt might write the first term as $\sec(x)\,\sin(x)^2$, which is the same thing. The answer $\sin^2 x+\sin x$ drops a denominator, the $+1$ version says $\tan x \cos x=1$, and $(\sin x)/(\cos x)\cdot\sin x+\cos x$ multiplies only the first term. **$(\sin^2 x)/(\cos x)+\sin x$.**

Example 6: Simplify 2

$$\csc x\,(\tan x - \cos x)$$

Question

Which expression equals this one?

- **A.** $\sec x + \cot x$
- **B.** $\sec x - 1$
- **C.** $\sec x - \cot x$
- **D.** $\csc x \tan x - \cos x$

Answer and reasoning

$\csc x \tan x - \csc x \cos x = 1/(\sin x)\cdot(\sin x)/(\cos x) - 1/(\sin x)\cdot\cos x = 1/(\cos x) - (\cos x)/(\sin x) = \sec x - \cot x$. CheckIt might leave the second term as $\cos(x)/\sin(x)$. The answer $\sec x - 1$ says $\csc x \cos x = 1$, the plus version loses the minus sign, and $\csc x \tan x - \cos x$ multiplies only the first term. **$\sec x - \cot x$.**

Example 7: Simplify 3

$$(\sin x+\cos x)\sec x\csc x$$

Question

Which expression equals this one?

- **A.** $\tan x+\cot x$
- **B.** $\sec x+\csc x$
- **C.** 2
- **D.** $\sin x\sec x+\cos x$

Answer and reasoning

$\sec x\csc x=1/(\sin x\cos x)$, so the expression is $(\sin x+\cos x)/(\sin x\cos x)=(\sin x)/(\sin x\cos x)+(\cos x)/(\sin x\cos x)=1/(\cos x)+1/(\sin x)$. Split the fraction into two, then each one simplifies. The answer $\tan x+\cot x$ keeps the wrong function on top, 2 assumes everything cancels, and $\sin x\sec x+\cos x$ skips the $\csc x$ on the second term. **$\sec x+\csc x$.**

Example 8: Simplify 4

$$(\tan x - \cot x)\sec x \csc x$$

Question

Which expression equals this one?

- **A.** $\sec x - \csc x$
- **B.** $\csc^2 x - \sec^2 x$
- **C.** $\tan x - \cot x$
- **D.** $\sec^2 x - \csc^2 x$

Answer and reasoning

Multiply each term by $1/(\sin x \cos x)$: $(\sin x)/(\cos x) \cdot 1/(\sin x \cos x) = 1/(\cos^2 x)$ and $(\cos x)/(\sin x) \cdot 1/(\sin x \cos x) = 1/(\sin^2 x)$. So the expression is $\sec^2 x - \csc^2 x$. The answer $\sec x - \csc x$ loses the squares, $\csc^2 x - \sec^2 x$ reverses the order, and $\tan x - \cot x$ never multiplies by $\sec x \csc x$. By the way, $\tan^2 x - \cot^2 x$ would also be correct, since $\tan^2 x = \sec^2 x - 1$ and $\cot^2 x = \csc^2 x - 1$. **$\sec^2 x - \csc^2 x$.**

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Example 9: Proof 1

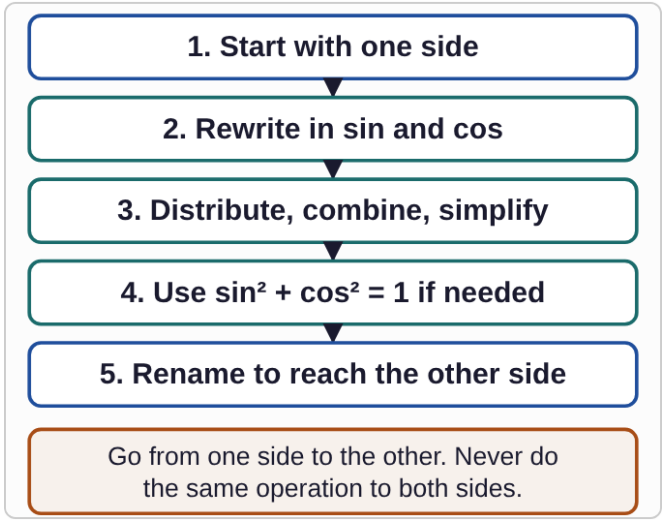


Figure: A flowchart with five numbered steps from top to bottom: 1, start with one side; 2, rewrite in sin and cos; 3, distribute, combine, simplify; 4, use sin squared plus cos squared equals 1 if needed; 5, rename to reach the other side. A box at the bottom says: go from one side to the other, never do the same operation to both sides.

$$\begin{aligned}&\cot x\,(\sin x - \sec x)\\ =&((\cos x)/(\sin x))(\sin x - 1/(\cos x))\\ =&\ ?\\ =&\cos x - \csc x\end{aligned}$$

Question

Which line fills the missing step?

- **A.** $\cos x - 1/(\cos x)$
- **B.** $1 - 1/(\sin x)$
- **C.** $\cos x - \cot x$
- **D.** $\cos x - 1/(\sin x)$

Answer and reasoning

Distribute: $(\cos x)/(\sin x)\cdot\sin x=\cos x$, and $(\cos x)/(\sin x)\cdot 1/(\cos x)=1/(\sin x)$. That gives $\cos x-1/(\sin x)$, and renaming $1/(\sin x)=\csc x$ finishes. The answer $\cos x-1/(\cos x)$ cancels the wrong factor, $1-1/(\sin x)$ says $\cot x \sin x=1$, and $\cos x-\cot x$ multiplies the second term by $(\cos x)/(\sin x)$ without cancelling. **$\cos x-1/(\sin x)$.**

Example 10: Proof 2

$$\begin{aligned}
&\sec x - \cos x \\
&= 1/(\cos x) - (\cos^2 x)/(\cos x) \\
&= (1 - \cos^2 x)/(\cos x) \\
&= \ ? \\
&= \sin x \cdot (\sin x)/(\cos x) \\
&= \sin x \tan x
\end{aligned}$$

Question

Which line fills the missing step?

- **A.** $(\sin x)/(\cos x)$
- **B.** $(\sin^2 x)/(\cos x)$
- **C.** $(\cos^2 x)/(\cos x)$
- **D.** $1/(\cos x) - 1$

Answer and reasoning

By the Pythagorean identity, $1 - \cos^2 x = \sin^2 x$, so the fraction is $(\sin^2 x)/(\cos x)$. Splitting $\sin^2 x$ into $\sin x \cdot \sin x$ gives the next line. The answer $(\sin x)/(\cos x)$ loses the square, $(\cos^2 x)/(\cos x)$ uses the wrong half of the identity, and $1/(\cos x) - 1$ cancels the cosine out of part of a difference. **$(\sin^2 x)/(\cos x)$.**

Example 11: Proof 3

$$\begin{aligned}
&\tan x+\cot x\\
&=(\sin x)/(\cos x)+(\cos x)/(\sin x)\\
&=\ ?\\
&=1/(\sin x\cos x)\\
&=\sec x\csc x
\end{aligned}$$

Question

Which line fills the missing step?

- **A.** $(\sin x+\cos x)/(\sin x\cos x)$
- **B.** $(\sin^2 x+\cos^2 x)/(\sin x+\cos x)$
- **C.** $(\sin^2 x+\cos^2 x)/(\sin x\cos x)$
- **D.** $(\sin x\cos x)/(\sin x\cos x)$

Answer and reasoning

The common denominator is $\sin x\cos x$. Multiply the first fraction by $(\sin x)/(\sin x)$ and the second by $(\cos x)/(\cos x)$: $(\sin^2 x+\cos^2 x)/(\sin x\cos x)$. The top is 1, which gives the next line. The answer $(\sin x+\cos x)/(\sin x\cos x)$ adds the tops without rewriting them, and the one with $\sin x+\cos x$ on the bottom adds the bottoms. **$(\sin^2 x+\cos^2 x)/(\sin x\cos x)$.**

Example 12: Proof 4

$$x=\pi/6$$

Question

Three of these equations are true for every x where both sides are defined. Which one is not? Testing the value above can help, or rewrite each one in sines and cosines.

A. $\sin x \sec x=\tan x$

B. $\tan x \cos x=\cos x$

C. $\cos x \csc x=\cot x$

D. $\sec x \cot x=\csc x$

Answer and reasoning

$\tan x \cos x=(\sin x)/(\cos x)\cdot\cos x=\sin x$, which is not $\cos x$. At $x=\pi/6$, the left side is $1/2$ and the right side is $(\sqrt{3})/2$. Careful: at $x=\pi/4$ both sides are $(\sqrt{2})/2$, so one lucky value proves nothing. A number can show an equation is false; only a verification shows it is true. The other three check out in sines and cosines. **$\tan x \cos x=\cos x$ is the false one.**

Example 13: Slip 1

$$(\sin x+\cos x)/(\sin x)=\cos x$$

Question

A student simplified this fraction by crossing out sin x on the top and the bottom. What is the mistake, if any?

A. There is no mistake.
B. It should be 1 + cos x, keeping the 1 from sin x over sin x.
C. Only part of the top is sin x, so it is 1 + cot x.
D. The cos x should be crossed out instead, leaving sin x.

Answer and reasoning

Only a factor of the whole top can be crossed out. Split the fraction instead: $(\sin x)/(\sin x)+(\cos x)/(\sin x)=1+\cot x$. Check at $x=\pi/4$: the fraction is 2, but $\cos(\pi/4)\approx 0.707$. **It equals $1+\cot x$.**

Example 14: Slip 2

$$
\begin{aligned}
&(\cos x - \sin x)\sec x \csc x \\
&= (\cos x - \sin x)/(\sin x \cos x) \\
&= 1/(\sin x) - 1/(\cos x) \\
&= \csc x - \sec x
\end{aligned}
$$

Question

A student verified this identity. What is the mistake, if any?

A. There is no mistake.
B. The last line should be sec x − csc x.
C. The first fraction cannot be split into two.
D. sec x csc x should be sin x cos x, so it is $\cos^2 x - \sin^2 x$.

Answer and reasoning

$\sec x \csc x = 1/(\sin x \cos x)$. Splitting gives $(\cos x)/(\sin x \cos x) - (\sin x)/(\sin x \cos x) = 1/(\sin x) - 1/(\cos x)$, and renaming gives $\csc x - \sec x$. Each line equals the one before, starting from one side and ending at the other. **There is no mistake.**

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Example 15: Slip 3

$$\sec x - \cos x = \sin x \tan x$$
$$1 - \cos^2 x = \sin^2 x$$
$$\sin^2 x = \sin^2 x$$

Question

To verify $\sec x - \cos x = \sin x \tan x$, a student multiplied both sides by $\cos x$ and simplified. What is the mistake, if any?

A. There is no mistake.

B. It should multiply both sides by sec x instead of by cos x.

C. $1 - \cos^2 x$ is $\cos^2 x$, not $\sin^2 x$.

D. Changing both sides needs a reason the steps reverse.

Answer and reasoning

The algebra is correct. But the student started from the identity itself, which is what is to be shown, and changed both sides. Ending at $\sin^2 x = \sin^2 x$ proves the identity only if every step can be reversed. Here it can: on the identity's domain $\cos x \neq 0$, so multiplying by $\cos x$ can be undone. The work is incomplete, not wrong, until it says so. The cleaner way, and the one CheckIt expects, is one chain from one side to the other: $\sec x - \cos x = (1 - \cos^2 x)/(\cos x) = (\sin^2 x)/(\cos x) = \sin x \tan x$. Multiplying by $\sec x$ would not help; and $1 - \cos^2 x = \sin^2 x$, from $\sin^2 x + \cos^2 x = 1$. **Justify that each step can be undone, or work from one side to the other.**

Example 16: Slip 4

$$-\sec x\,(\cos x - \cot x) = -1 - \csc x$$

Question

A student distributed and simplified this expression. What is the mistake, if any?

A. sec x times cos x is 0, so the answer is just −csc x.
B. The minus multiplies both terms, so it is −1 + csc x.
C. sec x times cot x is $\sec^2 x$, so it is $-1 - \sec^2 x$ instead.
D. There is no mistake.

Answer and reasoning

$-\sec x \cos x = -1$, and $-\sec x \cdot (-\cot x) = +1/(\cos x) \cdot (\cos x)/(\sin x) = +\csc x$. The negative in front changes the sign of every term inside. **It equals $-1+\csc x$.**

Example 17: Slip 5

$$(\sec x)/(\tan x)=(1/\sin x)/(\sin x/\cos x)=(\cos x)/(\sin^2 x)$$

Question

A student rewrote this quotient in sines and cosines. What is the mistake, if any?

A. sec x is 1/cos x, so the quotient is csc x.
B. The denominator should be sin x cos x, not $\sin^2 x$.
C. There is no mistake.
D. Dividing should flip the top fraction, giving $\sin^2 x/\cos x$.

Answer and reasoning

Secant goes with cosine: $\sec x=1/(\cos x)$, not $1/(\sin x)$. The student did divide correctly by flipping the bottom fraction. Then $(1/\cos x)/(\sin x/\cos x)=1/(\cos x)\cdot(\cos x)/(\sin x)=1/(\sin x)=\csc x$. **$(\sec x)/(\tan x)=\csc x$.**